space essential to achieve this goal? Robotic missions, given high national priority and equipped with improved machine intelligence, seem to me entirely capable of answering, as well as astronauts can, all the questions we need to ask—and at maybe 10 percent the cost.

It is alleged that "spinoff" will transpire—huge technological benefits that would otherwise fail to come about—thereby improving our international competitiveness and the domestic economy. But this is an old argument: Spend $80 billion (in contemporary money) to send *Apollo* astronauts to the Moon, and we'll throw in a free stickless frying pan. Plainly, if we're after frying pans, we can invest the money directly and save almost all of that $80 billion.

The argument is specious for other reasons as well, one of which is that DuPont's Teflon technology long antedated *Apollo*. The same is true of cardiac pacemakers, ballpoint pens, Velcro, and other purported spinoffs of the *Apollo* program. (I once had the opportunity to talk with the inventor of the cardiac pacemaker, who himself nearly had a coronary accident describing the injustice of what he perceived as NASA taking credit for his device.) If there are technologies we urgently need, then spend the money and develop them. Why go to Mars to do it?

Of course it would be impossible for so much new technology as NASA requires to be developed and not have some spillover into the general economy, some inventions useful down here. For example, the powdered orange juice substitute Tang was a product of the manned space program, and spinoffs have occurred in cordless tools, implanted cardiac defibrillators, liquid-cooled garments, and digital imaging—to name a few. But they hardly justify human voyages to Mars or the existence of NASA.

We could see the old spinoff engine wheezing and puffing in the waning days of the Reagan-era Star Wars office. Hydrogen bomb-driven X-ray lasers on orbiting battle stations will help perfect laser surgery, they told us. But if we need laser surgery, if it's a high national priority, by all means let's allocate the funds to develop it. Just leave Star Wars out of it. Spinoff justifications constitute an